

WARFARIN EDUCATION

ANDREW TRAN, PHARM.D

CLINICAL PHARMACIST

BAYLOR SCOTT & WHITE HEALTH – TEMPLE ANTICOAGULATION CLINIC

LEARNING OBJECTIVES

- After this presentation, the audience will be able to:
 - Understand the mechanism of action of warfarin
 - Explore dosing and management considerations of warfarin
 - Recognize drug, disease, and dietary interactions with warfarin

WARFARIN

1320353

Store at controlled room temperature (59°-86°F, 15°-30°C).

PROTECT FROM LIGHT AND MOISTURE.
Dispense in a tight, light-resistant container as defined in the USP.
RESEAL CAP TIGHTLY.

5mg

NDC 0056-0172-70
Rx only

COUMADIN®
(Warfarin Sodium Tablets, USP)
Crystalline*

DISPENSE WITH MEDICATION GUIDE

HIGHLY POTENT ANTICOAGULANT
WARNING: Serious bleeding results from overdosage. Do not use or dispense before reading directions and warnings in accompanying product information.

DOSAGE: See package insert.

100 TABLETS

*Present as crystalline sodium warfarin isopropanol dihydrate.
Distributed by: Bristol-Myers Squibb Company
Princeton, New Jersey 08543 USA

3 0056 0172 70 5

NDC 0832-1216-00

JANTOVEN®
Warfarin Sodium
Tablets, USP

5 mg

PHARMACIST: Dispense the Medication Guide provided separately to each patient.

HIGHLY POTENT ANTICOAGULANT WARNING:
Serious bleeding results from overdosage. Do not use or dispense before reading directions and warnings in accompanying product information.

100 Tablets

Rx only

UPSHER-SMITH

Each tablet contains: Warfarin Sodium 5 mg

Usual Adult Dosage: See package insert for full prescribing information.

Note to authorized dispenser: Obtain a Medication Guide by visiting www.upsheer-smith.com or by calling 1-888-650-3789.

Store at 20° to 25°C (68° to 77°F). Excursions permitted to 15° to 30°C (59° to 86°F). [See USP Controlled Room Temperature.] Keep tightly closed. Protect from light and moisture. Dispense in a tight, light-resistant container with a child-resistant closure.

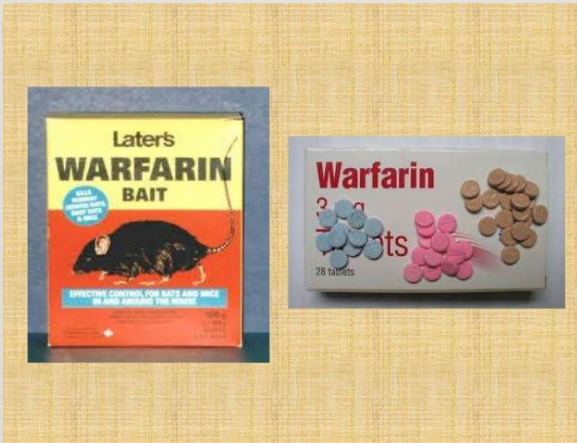
Keep out of reach of children.

SEALED FOR YOUR PROTECTION
Manufactured by
UPSHER-SMITH LABORATORIES, LLC
Maple Grove, MN 55369
© 2017 Upsheer-Smith Laboratories, LLC
112614-01 R0617

3 0832-1216-00 2

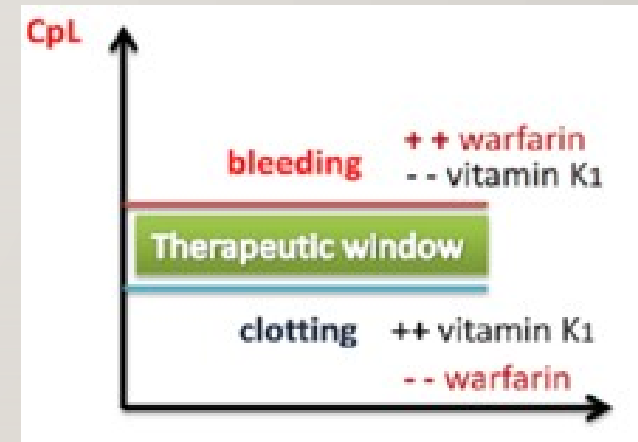
HISTORY OF WARFARIN

- One of oldest blood thinners
 - Early 1920s – Hemorrhagic events occurred in cattle that consumed spoiled sweet clover
 - 1940 – University of Wisconsin isolated the compound called 4-hydroxycoumarin
 - **W**isconsin **A**lumni **R**esearch **F**oundation (WARF) + **-ARIN** (from coumarin).
 - Developed as rat poison
- medical use in humans in 1954



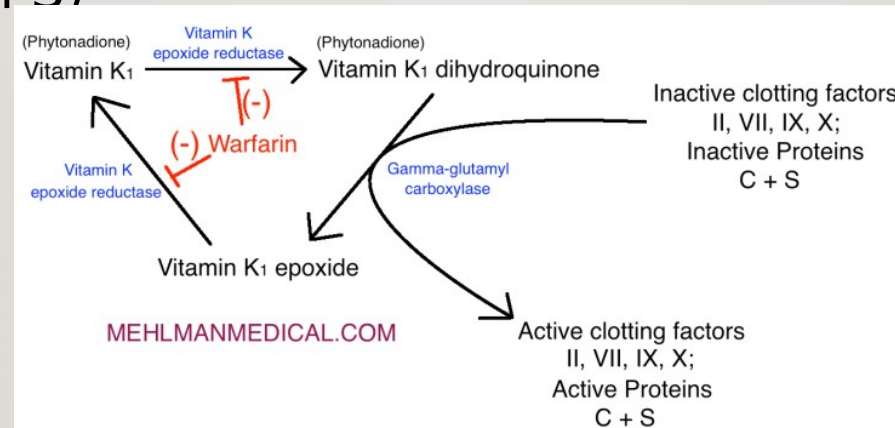
WARFARIN

- Narrow therapeutic index; changes frequently
- Recently fallen out of favor compared to DOACs (Eliquis, Xarelto) for many conditions
- Still drug of choice:
 - Valvular Disease (mechanical valve replacement)
 - Antiphospholipid syndrome
 - LVAD

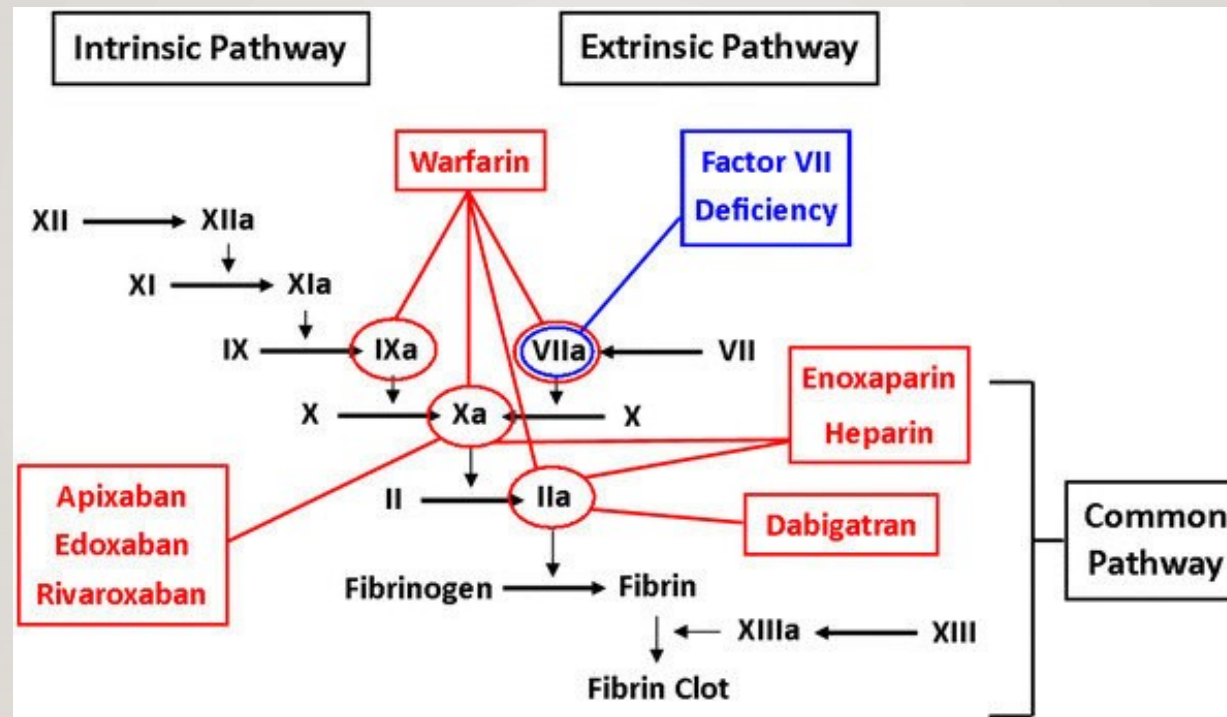


HOW DOES WARFARIN WORK?

- Mechanism of action - inhibits the enzyme that is responsible for conversion of vitamin K in the liver.
- Vitamin K is an important for several clotting factors (II, VII, IX, X; Protein C, Protein S)



CLOTTING CASCADE



PT/INR

RESULTS IN
≈5 MINUTES



PT/INR - EXAMPLE

PROTIME W/ INR	 	
Prothrombin Time (PT)		
Prothrombin Time (PT)	27.4 ▲ 	
International Normalized Ratio (INR)	2.5 ▲ 	

MONITORING

- PT/INR
 - Prothrombin time – how long it takes for plasma to clot after adding tissue factor + calcium (seconds)
 - International normalized ratio:

$$INR = \left(\frac{PT_{\text{test}}}{PT_{\text{normal}}} \right)^{ISI}$$

- **Patient's PT** = The **prothrombin time** (in seconds) measured in the patient's blood sample.
- **Mean Normal PT** = The **average PT of a normal control sample** (determined by the lab).
- **ISI (International Sensitivity Index)** = A correction factor that accounts for differences in thromboplastin reagents used in PT tests.

PT/INR GOALS

- INR: ↑ blood thinner/more warfarin in body; ↓ blood thicker/less warfarin in body
- Baseline INR = 0.8-1.1
- Target INR is usually 2.5 (range 2-3) for most conditions (PE/DVT); 3.0 (range 2.5-3.5) for mechanical valve replacement

PROTIME W/ INR	
Prothrombin Time (PT)	
Prothrombin Time (PT)	27.4 ▲ 📄
International Normalized Ratio (INR)	2.5 ▲ 📄

CUSTOM INR RANGES

- Custom INR ranges can be used for patient specific factors (high bleeding/clotting risk; LVAD; etc)
- Example: INR 1.8-2.2 in some LVAD patients

MAIN TEXT

Artificial
Organs  WILEY

Evaluation of adjusted international normalized ratio goal in patients with HeartMate 3 left ventricular assist devices

Amanda Sowder¹  | Ayesha Ather^{1,2} | Emma Birks^{1,3} | Andrew Kolodziej^{1,3} | Rajasekhar Malyala^{1,3} | Adam Sieg^{1,2} | Aric Schadler^{1,2} | Rachel Unger¹ | Maya Guglin⁴ | William Kuan^{1,2}

¹Department of Cardiology, University of Kentucky Healthcare, Lexington, Kentucky, USA

²Department of Pharmaceutical Sciences, University of Kentucky College of Pharmacy, Lexington, Kentucky, USA

³Department of Cardiology, University of Kentucky College of Medicine, Lexington, Kentucky, USA

⁴Department of Cardiology, Indiana University Health, Indianapolis, Indiana, USA

Correspondence

Amanda Sowder, AdventHealth Orlando, 601 E. Rollins Street, Orlando, FL 32803, USA.
Email: amanda.sowder@adventhealth.com

Abstract

Background: The HeartMate3 left ventricular assist device (HM3 LVAD) has shown a low incidence of thrombosis, but bleeding risk is as high as 43%. We aim to describe the impact of lower international normalization ratio (INR) goal on clinical outcomes.

Methods: In February 2019, our tertiary care institution lowered INR goal in HM3 patients from manufacturer recommendations to **1.8–2.2** and retrospectively analyzed the data. Two cohorts were compared: patients with lower INR goal upon implant (De novo) and those with subsequently lowered INR goal (Adjusted). The Adjusted group also served as its own historical control. Both groups continued aspirin 81 milligrams daily per manufacturer recommendations. The primary outcomes were incidences of bleed and thrombosis events within 12 months. Secondary outcomes included survival free of disabling stroke or reoperation to remove or replace the device and Rosendaal time in therapeutic range (TTR) over 12 months.

WARFARIN - DOSING

Coumadin (warfarin tablets)



Warfarin tablets (Barr brand)



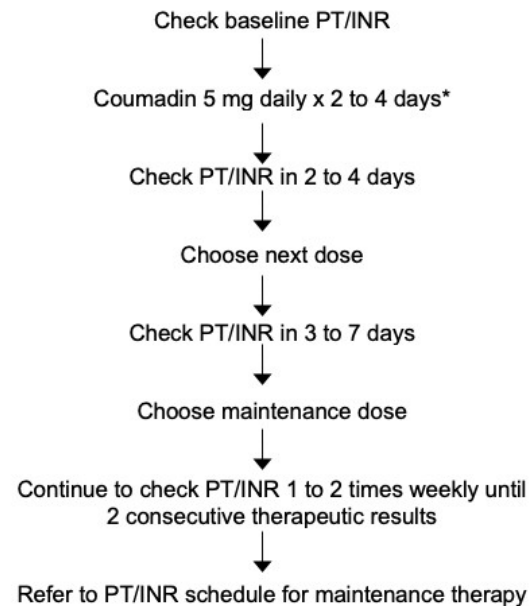
WARFARIN - DOSING

- Warfarin tablets have a color that indicates their strength, or how many milligrams (mg) are in each tablet.
- CHEST Guidelines: start 10 mg daily x 2 days then adjust
- In practice: typical to start at 5 mg daily; may start 2.5 mg daily (elderly)



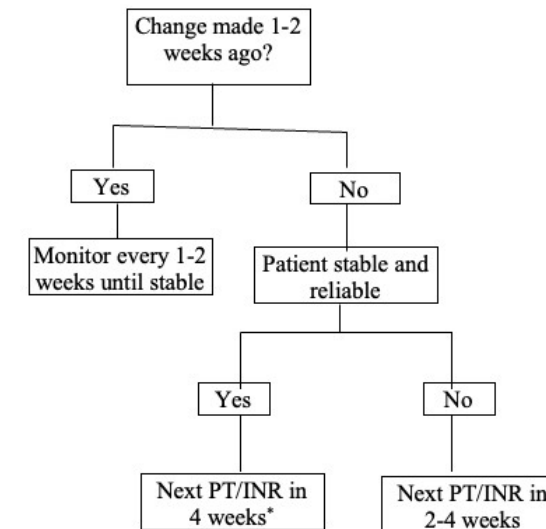
EXAMPLE - DOSING PROTOCOL

PROTOCOL FOR INITIATION OF COUMADIN THERAPY



* Initial doses vary up to 10 mg; suggested initial dose in the elderly is ≤ 5 mg.

PT/INR SCHEDULE FOR MAINTENANCE THERAPY



* For patients who have been stable for at least one month, the next PT/INR may be deferred for 4 weeks.

EXAMPLE – DOSING SCHEDULES

Warfarin Dosing Schedule

Mon	Tue	Wed	Thu	Fri	Sat	Sun	Total Weekly Dose
5	5	5	5	5	5	5	35 mg
2.5	5	5	2.5	5	5	5	30 mg
2.5	5	2.5	5	2.5	5	5	27.5 mg

WARFARIN - INTERACTIONS



WARFARIN - DRUG-DISEASE STATE INTERACTIONS

- ↑ INR

- Advanced age
- Binge drinking
- Liver disease
- Heart Failure
- Kidney Disease
- Hyperthyroidism (↑ breakdown of clotting factors)
- Acute Infection or inflammation
- Fever
- Malignancy
- Diarrhea

- ↓ INR

- Chronic alcoholism (variable)
- Smoking/tobacco use
- Hypothyroidism (↓ breakdown of clotting factors)
- HIGH protein diet (↑ increased production of albumin)

WARFARIN - DRUG INTERACTIONS – MANY!

- ↑ INR

- Amiodarone
- Bactrim®
(sulfamethoxazole/trimethoprim)
- Fluconazole (Diflucan®)
- Metronidazole (Flagyl®)
- Other antibiotics
- Alcohol

- ↓ INR

- Vitamin K
- Rifampin
- Phenytoin
- Smoking/Tobacco

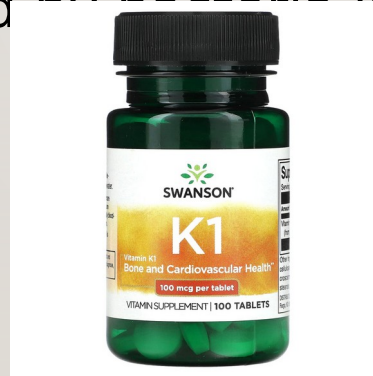
VITAMIN K

- More Vitamin K: ↓ INR
- Less Vitamin K: ↑ INR
- Counsel patients to stay consistent with vitamin K; eat the same amount of Vitamin K daily/weekly

Warfarin	Vitamin K
helps to "thin blood" and prevent blood clots	helps blood to clot
	

VITAMIN K

- Two types: Vitamin K₁ and K₂
- K1: is primarily found in **dark green leafy vegetables** (e.g., spinach, kale, broccoli)
- K2: found in animal products (e.g., fermented foods; cheese; organ meats) and produced by bacteria in the gut
- BOTH affect INR.



VITAMIN K FOODS



Low (<50 mcg)	Moderate (50-100mcg)	High (>100mcg)
Vegetables Alfalfa sprouts (1½ cup) Artichoke (1 medium; 1 cup) Beans, green (2 cup) Beans, lima, kidney (>2 cup) Beans, navy (>2 cups) Beans, yellow (2 cup) Beats (>2 cups) Carrots (2 cup) Cauliflower (2 cup) Celery (2½ sticks) Corn (½ cup) Cucumber (2 cup) Egg plant (1¼ cup) Garbanzo beans (½ cup) Mushrooms (1½ cup) Okra (½ cup) Onion Peas, blackeyed (1 cup) Peas, green (1 cup) Pepper, green (½ cup) Potato , Pumpkin Sauerkraut (½ cup) Soybeans (¾ cup) Tomato, green (½ cup) Tomato, red (¾ cup)	Asparagus (7-12 spears) Avocado (½ cup) Peas, green (1½ cup) Lettuce (iceberg – ½ head) Pickle, dill (1 medium) Spinach (1/4 cup – raw)	Beet greens (¼ cup) Broccoli (½ cup) Brussels sprouts (½ cup) Cabbage (¾ cup) Collard greens (½ cup) Endive (1 cup) Green Scallions (½ cup) Kale (½ cup) Mustard greens (1 cup) Parsley (½ cup) Spinach (¼ cup - cooked) Swiss chard (¼ cup) Turnip greens (¼ cup) Watercress (raw chopped) – 1¼ cup
Fruits Apple, Apricot, Banana, Blueberries (< 1 cup) Cantaloupe, Lemon, Orange, Peach, Pear, Plum, Pineapple	Grapes (2 ½cups)	
Meats Beef, Chicken, Ham, Mackerel Pork, Shrimp, Turkey	Tuna (> 4oz)	Liver
Fats and Oils Canola oil (< 3 Tbspn) Olive oil (< 6 Tbspn) Corn oil, Peanut oil, Safflower oil, Sesame oil, Sunflower oil (all < 7 Tbspn)	Margarine (4 Tbspn) Mayonnaise (>10 Tbspn) Soybean oil (2-4 Tbspn)	Canola oil (>6 Tbspn)
Dairy products Butter ,Cheese, Eggs, Sour cream Yogurt		
Beverages Coffee, Cola, Fruit juices, Milk, Tea (black)		Tea (green): Brewing green tea may alters vitamin K content
Grains, breads Bagel, Breads, Cereal, Flour, Oatmeal, Rice, Spaghetti		
Misc. Honey, Jell-O Gelatin Peanut butter, Sugar, Tofu		

WARFARIN - HERBAL INTERACTIONS – MANY!

- Monitor for interactions with herbal supplements
- ↑ INR examples
 - "5Gs" – garlic, ginger, ginkgo, ginseng, glucosamine
 - Grapefruit
 - Cranberry supplements
- ↓ INR examples
 - Green Tea
 - Coenzyme Q10

Warfarin (Coumadin®) ↔ Herbal Interactions

Increased Risk of Clotting	Increased Risk of Bleeding
----------------------------	----------------------------

High Protein Diet
Coenzyme Q10
Green Tea
St John's Wort
Vitamin C
Vitamin K
Alfalfa

Acai
Cranberry
Grapefruit
Noni
Pomegranate
Garlic
Licorice
Willow Bark
Boldo-fenugreek
Danshen
Dong quai
Feverfew
Fish Oil
Ginger
Ginkgo
Orlistat
PC-SPEs
Guilinggao (turtle jelly)
Vitamin E
Glucosamine
Chondroitin
Kava
Melatonin
Saw Palmetto
Policosanol



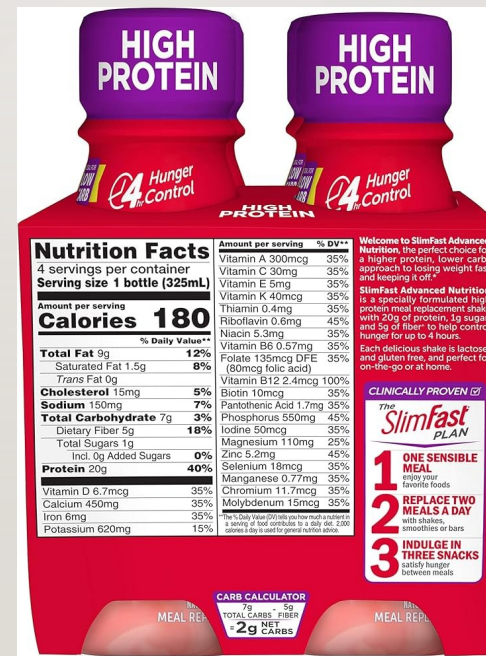
Many herbal products can change how warfarin (Coumadin) works.

Some of the most commonly seen interactions are listed above

Many prescriptions and over-the-counter medications also interact with warfarin.

WARFARIN – OTHER CONSIDERATIONS

- Counsel patient to stay consistent and report changes in vitamins and nutritional drinks!



WARFARIN - OTHER CONSIDERATIONS

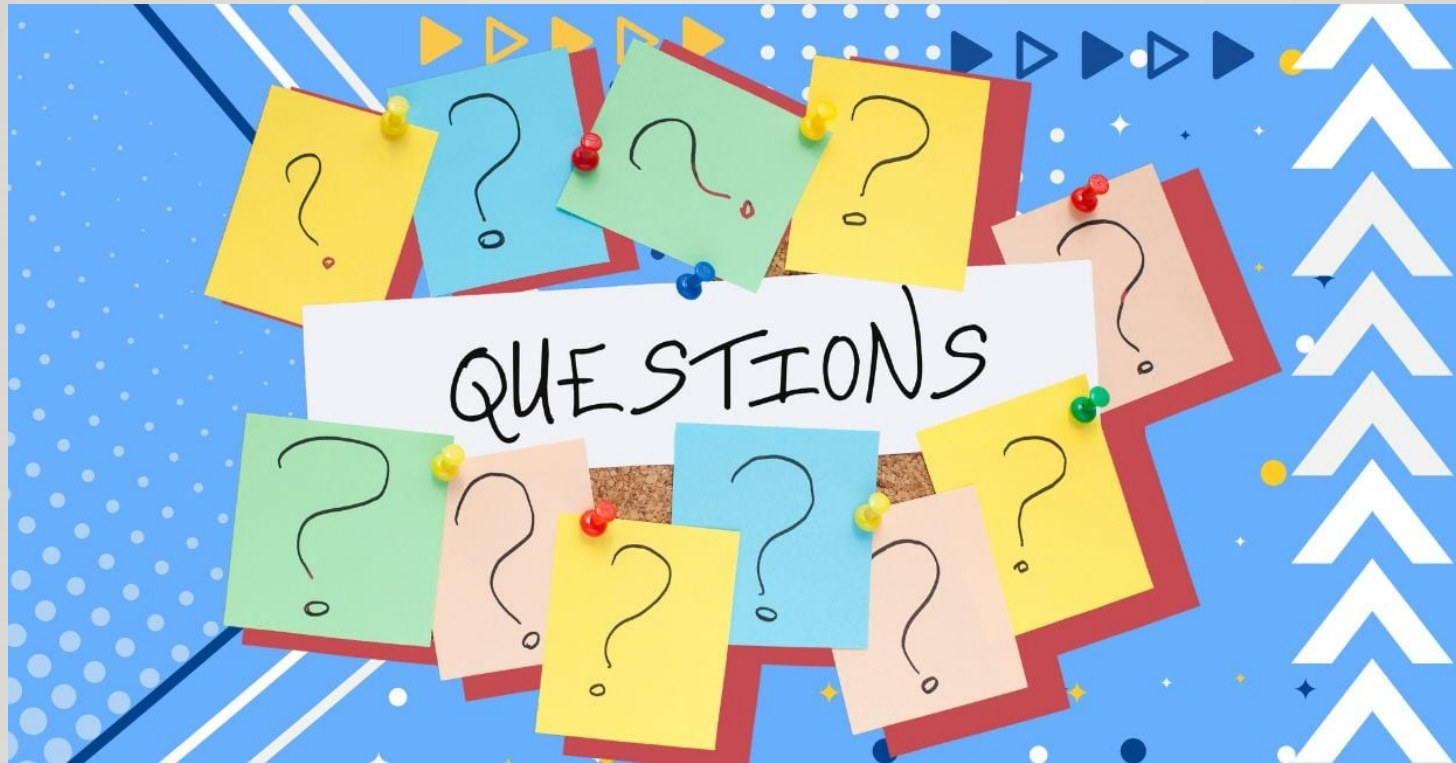
- Over-the-counter pain medications-
 - Avoid NSAIDs (e.g. Advil or Motrin (Ibuprofen); Aleve (Naproxen); aspirin)
 - ↑ GI bleeding
 - Tylenol/acetaminophen is PREFERRED.
 - Note: max dose for warfarin patients: 2000 mg/day (usual max dose: 4000



CONCLUSION

- Counsel patients to stay **consistent** in vitamin K intake (diet, vitamins, nutritional drinks, etc.).
- Advise patients to report changes in diet, medications, supplements, smoking/drinking, and overall health.
- Patient may need more frequent monitoring with bigger changes.

QUESTIONS?



WARFARIN EDUCATION

ANDREW TRAN, PHARM.D

EMAIL: ANDREW.TRAN@BSWHEALTH.ORG